

Nikhil Rajan

n.rajan2@lse.ac.uk | [LinkedIn](#) | [Personal Website](#) | London, UK

EDUCATION

London School of Economics and Political Science

MSc in Financial Statistics

London, UK

Sep 2026 – Jun 2027

- **Coursework:** Stochastic Processes, Mathematics of Black & Scholes Theory, Fixed Income, Mathematics of market microstructure, Stochastic Simulation and Calibration, Financial Statistics 1 & 2 (Machine Learning), Advanced Statistics.

Madras School of Economics

Bachelor of Arts (Honours) in Economics

Chennai, India

Aug 2023 – Jun 2026

- **Cumulative Grade Point Average:** 9.61/10.0 | **Gold Medallist (Rank 1/70).**
- **Dean's List (Distinction):** Semesters II–VI.
- **Coursework:** Stochastic Processes, Real Analysis, Advanced Time Series, Differential Equations, Linear Algebra, Econometrics, Statistics, Game Theory, Money and Banking, Data Science, Python and MATLAB.

Princeton University

Macro, Money & Finance Summer School

Online

May 2026 – Jul 2026

- Completed the Markus' Academy Ph.D.-level summer program run by Prof. Markus Brunnermeier, covering continuous-time macro-finance, stochastic calculus, and asset pricing.

EXPERIENCE

Yosaney Fintech Private Limited

Finance Intern

Chennai, India

May 2024 – Jul 2024

- Built automated tools generating P&L statements, balance sheets, and financial schedules that enabled real-time credit assessment for 100+ MSMEs seeking bank and NBFC financing.
- Analysed MSME financial statements and evaluated bank/NBFC credit products for documentation and risk requirements.
- Reduced each reporting cycle from roughly a full working day to about 1.5 hours (around 80% faster).

Madras School of Economics

Research Intern (Advisor: Dr. Naveen Srinivasan)

Chennai, India

Dec 2024 – Jan 2025

- Investigated horizontal innovation and knowledge spillovers in the Romer growth model using analytical and numerical methods.
- Proposed and evaluated approaches to address a research question linking institutional quality, innovation, and productivity.

QUANTITATIVE RESEARCH PROJECTS

Liquidity Regimes as a Markov Chain and Optimal Execution as a Markov Decision Process

2025 (Ongoing)

Modelling liquidity as a four-state Markov process and developing continuous-time optimal (algorithmic) trade-execution strategies under persistent liquidity regimes.

Causal Analysis of Jump Contagion in Equity Markets: Evidence from Nifty 50 and S&P 500 Returns

2026

Examined jump spillovers between S&P 500 and Nifty 50 returns, finding a 0.284 transmission coefficient and $3.42\times$ higher volatility during crisis periods.

Hidden Markov Model Framework for Credit Stress Detection and Recession Early Warning

2026 (Ongoing)

Developing a hidden Markov model and mean-reverting process to identify credit-stress regimes, link them to recessions, and build a backtested, systematic portfolio-allocation strategy.

Credit Rating Transitions and Markov Chains

2025

Reconstructed the Jarrow-Lando-Turnbull (1997) credit migration model from first principles and derived hedging strategies and risk-neutral transition matrices.

Stochastic Welfare Analysis of Sin Taxation Under India's New GST Regime

2024

Demonstrated that cigarette tax hikes increase household welfare in two Indian states using Cobb-Douglas utility and Markov chains.

TEACHING EXPERIENCE

Madras School of Economics

Teaching Assistant, Quantitative Courses

Chennai, India

Mar 2024 – May 2026

- Led tutorials for six quantitative courses over two years: Stochastic Processes, Time Series, Game Theory, Multivariable Calculus, Intermediate Microeconomics, and Foundations of Modern Finance.
- Taught up to two quantitative courses concurrently in a single term, leading weekly tutorials and problem-solving sessions.
- Authored lecture notes, quizzes, and assignments spanning martingales, Markov decision processes, Brownian motion, CAPM, stochastic discount factors, and derivatives pricing.

SKILLS

- **Programming & Tools:** Python, R, MATLAB, Stata, Excel, LaTeX.
- **Quantitative Finance:** Derivatives & Option Pricing, Credit Risk Modelling, Asset Pricing & Valuation, Portfolio Optimisation, Optimal Execution, Market Microstructure.
- **Modelling & Methods:** Stochastic Modelling, Monte Carlo Simulation & Calibration, Time-Series & Regime Detection, Machine Learning, Backtesting, Numerical Methods.

EXTRACURRICULAR ACHIEVEMENTS

Chess

FIDE-rated Player

- Placed 13th at the 1st SRM International Open (FIDE Rapid), 2022.
- Captained the school team; scored 5.5/6 on Board 1 at the CBSE Cluster Championships, 2022.